

Lab-Scale Multi-Stage Packed-Bed Water Filtration Column

1. Executive Summary & Overview

This report documents the design, construction, and performance evaluation of a laboratory-scale 4-stage packed-bed water filtration column. The objective of this project was to analyze how the fluid moves throughout the system, volumetric flow rates (Q), liquid volume left, and depth filtration efficiency using distinct filtering layers.

2. Column Architecture & Technical Specifications

The column was constructed in an inverted cylindrical vessel with a mesh retention cap at the outlet to ensure material stability and prevent blowout.

Layer Position	Media Type	Grain / Particle Size	Primary Engineering Function
Top (Layer 1)	Coarse Gravel	5 – 10 mm	Energy dissipation, equalizing flow, and capturing large debris
Middle (Layer 2)	Fine Sand	0.5 – 1.0 mm	Finer straining, filtering with depth
Middle (Layer 3)	Activated Carbon	Extruded Pellets	Chemical adsorption of organic colorants (dyes) and dissolved solutes
Bottom (Layer 4)	Poly-Filter Cotton	Cotton Balls	Retention barrier

3. Experimental Procedure & Mass Balance Results

A consistent feed volume ($V_{in} = 250.0$ mL) was introduced across three trials. Outflow liquid volume (V_{out}) and total filtration elapsed time (t) were recorded to determine actual discharge volumetric flow rate ($Q = V_{out} / t$) and volumetric recovery yield.

Trial	V_{in} (mL)	V_{out} (mL)	Yield (%)	Time, t (s)	Flow Rate, Q (mL/s)	Outflow Clarity (Qualitative)
Run 1	250.0	220.0	88.0%	105.2	2.09	• Very cloudy, (residual carbon) • Fast outflow
Run 2	250.0	235.0	94.0%	113.4	2.07	• Very cloudy, (residual carbon) • Outflow slowed (especially initial)
Run 3	250.0	236.0	94.4%	114.7	2.06	• Highest clarity observed although residual carbon was still found (optimal) • Similar outflow speed to previous trial
Average	250.0	230.3	92.1%	111.1	2.07	Steady-state flow established

4. Discussion & Transport Phenomena Analysis

4.1 Liquid Holdup & Media Saturation

- **Porous Media Storage:** Run 1 showed a liquid retention loss of 30.0 mL (12.0%), attributed to the filling and wetting of the open space across the sand and carbon.
- **Steady-State Recovery:** By Run 3, liquid holdup stabilized at ~13.5 mL (5.4% loss), demonstrating that initial losses were driven by pore space saturation rather than system leakage.

4.2 Mat Filtration vs. Flow Resistance

- **Cake Formation:** Total filtration time increased by 9.5 seconds between Run 1 and Run 3. Suspended particles formed a secondary filter mat on top of the the fine sand layer, increasing the efficiency of particle retention (cleaner outflow) at the cost of the ease of flow
- **Discharge Velocity Stability:** Calculating flow rate based on actual output volume (V_{out}) revealed that true discharge velocity remained remarkably stable (~2.06 – 2.09 mL/s).

5. Engineering Troubleshooting & Design Iterations

1. **Blowout:** Downward hydraulic pressure initially threatened outlet plug stability. Solved by creating a mesh screw cap to maintain compaction while maintaining uniform drainage.
2. **Residue:** To maximize filter performance, the pre-trials will include flushing the filtration system with clean water to prevent residue exiting along with the liquid

6. Proposed Future Work

- Integration of a mesh membrane at the bottom outlet.
- Flushing the system as a pre-trial to prevent residue entering the outflow.